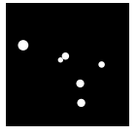
Module 2 - stage 1 pretraining step (train_module2_nonlong_pretrain.py)



X EyePACS image
(3ch, grade s)
+ Module 1
mask (1ch)



concat, 4+c_dim ch

Generator



G(x,c)
stage-c
image

Discriminator
(PatchGAN
+ grade head)

d_loss = fake − real
+ 10·GP + CE(D_cls(X), s)

g_loss = adv
+ CE(D_cls(G(x,c)), c)
+ 10·L1_cyc + 5·L1_id
+ 1·L_bio + 1·L_mono

40 epochs · 20,000
images per epoch

real X, graded with its own stage s

c target stage, one-hot
s+1 with p = 0.5,
else any stage

G(G(x,c), s)
cycle back to the
source stage

L1
cyc

X

Biomarker
regressor R

R(G(x,c)):
EX / HE / MA / SE
area + count,
LBS (9 values)

bio +
mono

stage profiles from
Module 1 on EyePACS
(stage_biomarker_
profiles.json)

R is trained on real images to predict Module 1's biomarkers; mono: lesion burden of G(x,c) ≥ burden of X when c > s